

Solving Quadratic Equations Using the Quadratic Formula

Jonathan R. Bacolod

Quadratic Equation

A **quadratic equation** is an equation that can be written in the standard form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

The Quadratic Formula

The **quadratic formula** is used to solve any quadratic equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Step 1: Standard Form

Before using the formula, write the equation in **standard form**.

$$ax^2 + bx + c = 0$$

Step 2: Identify a, b, c

Identify the values of **a**, **b**, and **c** correctly.

$$ax^2 + bx + c = 0$$

Step 3: The Discriminant

Compute the **discriminant**,

$$b^2 - 4ac$$

to determine the nature of the roots.

Case: Two Real Solutions

If

$$b^2 - 4ac > 0$$

the equation has **two distinct real solutions**.

Case: One Real Solution

If

$$b^2 - 4ac = 0$$

the equation has **one repeated real solution.**

Case: Two Complex Solutions

If

$$b^2 - 4ac < 0$$

the equation has **two complex solutions**.

Simplify and Verify

Simplify radicals whenever possible and write the final answers in simplest form.

Always **verify** the solutions by substituting them into the original equation when necessary.

Discriminant at a Glance

Discriminant	Nature of Roots
$b^2 - 4ac > 0$	Two distinct real solutions
$b^2 - 4ac = 0$	One repeated real solution
$b^2 - 4ac < 0$	Two complex solutions

Examples

Example 1: Two Real Solutions

Solve

$$x^2 - 3x - 4 = 0$$

Here $a = 1$, $b = -3$, $c = -4$.

$$b^2 - 4ac = 9 + 16 = 25$$

$$x = \frac{3 \pm 5}{2}$$

$$x = 4 \quad \text{or} \quad x = -1$$

Example 2: One Repeated Solution

Solve

$$x^2 - 4x + 4 = 0$$

Here $a = 1$, $b = -4$, $c = 4$.

$$b^2 - 4ac = 16 - 16 = 0$$

$$x = \frac{4}{2} = 2$$

Example 3: Two Complex Solutions

Solve

$$x^2 + 2x + 5 = 0$$

Here $a = 1$, $b = 2$, $c = 5$.

$$b^2 - 4ac = 4 - 20 = -16$$

$$x = \frac{-2 \pm 4i}{2} = -1 \pm 2i$$

Practice Exercises

Practice 1

Solve using the quadratic formula.

$$x^2 - 5x + 6 = 0$$

Practice 2

Solve using the quadratic formula.

$$2x^2 + 7x + 3 = 0$$

Practice 3

Solve using the quadratic formula.

$$3x^2 - 2x - 8 = 0$$

Practice 4

Solve using the quadratic formula.

$$5x^2 + x - 6 = 0$$

Practice 5

Solve using the quadratic formula.

$$4x^2 - 4x + 1 = 0$$

Practice 6

Solve using the quadratic formula.

$$x^2 + 6x + 13 = 0$$

Practice 7

Solve using the quadratic formula.

$$6x^2 - 5x - 6 = 0$$

Practice 8

Solve using the quadratic formula.

$$2x^2 - 8x + 5 = 0$$

Practice 9

Solve using the quadratic formula.

$$7x^2 + 2x - 3 = 0$$

Practice 10

Solve using the quadratic formula.

$$3x^2 + 5x + 1 = 0$$

Activities

Activity 1

Solve using the quadratic formula.

$$x^2 - 7x + 12 = 0$$

Activity 2

Solve using the quadratic formula.

$$2x^2 + 9x + 4 = 0$$

Activity 3

Solve using the quadratic formula.

$$4x^2 - 3x - 10 = 0$$

Activity 4

Solve using the quadratic formula.

$$3x^2 + 2x - 8 = 0$$

Activity 5

Solve using the quadratic formula.

$$9x^2 - 12x + 4 = 0$$

Activity 6

Solve using the quadratic formula.

$$x^2 + 8x + 20 = 0$$

Activity 7

Solve using the quadratic formula.

$$5x^2 - 7x - 6 = 0$$

Activity 8

Solve using the quadratic formula.

$$3x^2 - 10x + 6 = 0$$

Activity 9

Solve using the quadratic formula.

$$8x^2 + x - 2 = 0$$

Activity 10

Solve using the quadratic formula.

$$2x^2 + 7x + 2 = 0$$

Thank you!